

GRA-MEDIATED SYMBIOSIS OF HUMAN AND AI AGENT: FROM ONTOLOGICAL INCOMPATIBILITY TO SYNERGETIC HARMONIZATION

Abstract

This paper addresses the fundamental problem of integrating humans and artificial intelligence (or robots) into a unified cognitive-action system. We argue that direct “hybridization” is impossible due to ontological differences: humans are finite, historical, and subjective, whereas robots/AI agents are time-invariant, depend only on energy supply, and lack an internal temporal dimension. Nevertheless, we demonstrate that the GRA methodology (Godel–Reductio ad Absurdum) and its iterative self-optimization pipeline GRA–ISOC provide a formal apparatus for creating a symbiotic interface that allows both participants to derive mutual benefit without losing their essence. We propose a mathematical model for aligning subjective and objective representations through a harmonization functional, whose minimization ensures coherent interaction. Practical applications in medicine, education, scientific discovery, and complex system management are described. A protocol for in silico verification of symbiosis effectiveness is provided, along with implementation recommendations.

Keywords: GRA, GRA–ISOC, human–machine interaction, ontological incompatibility, symbiosis, harmonization, cognitive foam, consistency functional, in silico verification, negentropy.

1 Introduction: The Dilemma of Human–AI “Hybridization”

The rapid development of artificial intelligence and robotics raises the question of deep integration between humans and machines. However, attempts at direct “hybridization”—such as neurointerface implantation, complete replacement of cognitive functions, or merg-

ing of consciousness—encounter fundamental obstacles. Humans and robots (or AI agents) possess **different ontological statuses**:

- **Human**—a biological, finite, historical being, subject to aging, emotional, possessing subjective experience and free will.
- **Robot/AI agent**—a mechanical or virtual system, invariant with respect to time, functioning in the presence of energy and lacking internal temporality; its “life” is, in principle, infinite.

These differences make direct merging impossible without losing the essential properties of one party. However, this does not mean that cooperation and mutual complementation are excluded. On the contrary, the **GRA (Godel–Reductio ad Absurdum)** methodology and its pipeline implementation **GRA–ISOC** offer a formal method of *mediated harmonization*, in which humans and AI agents remain autonomous but exchange structured signals that eliminate contradictions and increase overall efficiency.

In this article, we show how the GRA approach allows turning the ontological abyss into a source of synergy, extracting benefits for both participants.

2 Ontological Incompatibility as Challenge and Resource

2.1 Robot Invariance vs. Human Finitude

A robot or AI agent can function indefinitely, provided there is continuous power supply and no physical damage. Its behavior is entirely determined by its program, which does not depend on biological time. Humans, on the other hand, have a finite life cycle, their cognitive abilities change with age, and their perception of time is deeply subjective.

This difference can be expressed in terms of *temporal operators*. Let T_H be the operator of evolution of the human system in time, and T_R the operator of evolution of the robot. Then:

$$T_H : S_H(t) \rightarrow S_H(t + \Delta t), \text{ taking into account aging, learning, emotions,}$$

$$T_R : S_R(t) \rightarrow S_R(t + \Delta t), \text{ where } S_R \text{ changes only by commands and external data.}$$

Moreover, T_R is reversible in many cases (state can be restored from a backup), whereas T_H is irreversible.

2.2 Why Direct “Hybridization” Is Impossible

Direct merging would require unification of these operators, which would mean either objectification of human consciousness (loss of subjectivity) or anthropomorphization of the machine (introduction of artificial mortality and irrationality), which destroys its useful properties—accuracy, reproducibility, infinite memory.

Thus, the task is not to “hybridize” them, but to *create an effective interface* that allows both to preserve their ontological characteristics while mutually enhancing each other.

3 GRA Approach to Human–AI Agent Harmonization

3.1 Basic Principles of GRA

The GRA methodology is based on measuring internal contradictions (cognitive foam) and their systematic elimination through iterative revision—*reductio ad absurdum*. A system is considered stable when the contradiction functional J becomes zero, and its first and second variations satisfy stability conditions.

In the context of human–machine symbiosis, we introduce **two levels**:

- **Subjective level** S_H —representation of the world in human consciousness (emotions, intuition, values).
- **Objective level** S_R —representation of the world in the AI agent (data, algorithms, probabilistic models).

Each level generates its own “foam”— J_H and J_R , respectively. In addition, there exists *cross-level foam* J_{HR} , arising from the mismatch between human expectations and machine inferences, as well as from human misunderstanding of AI logic and vice versa.

3.2 Harmonization Functional

We define the full harmonization functional Φ as:

$$\Phi = \alpha_H J_H(S_H) + \alpha_R J_R(S_R) + \alpha_{HR} d(S_H, L_{R \rightarrow H} S_R) + \alpha_{RH} d(S_R, L_{H \rightarrow R} S_H), \quad (1)$$

where:

- J_H —measure of internal contradictions in the human cognitive system (e.g., cognitive dissonance);

- J_R —measure of contradictions in the AI system (e.g., logical collisions in the knowledge base);
- $d(\cdot, \cdot)$ —metric of distance between representations;
- $L_{R \rightarrow H}$ —lifting operator, translating machine data into a human-perceivable form;
- $L_{H \rightarrow R}$ —lifting operator, translating human intentions and values into formal constraints for AI;
- α —weight coefficients.

The goal of the GRA step is to find S_H^* and S_R^* that minimize Φ :

$$(S_H^*, S_R^*) = \arg \min_{S_H, S_R} \Phi(S_H, S_R). \quad (2)$$

Here, neither S_H nor S_R merge; they remain independent, but their interaction becomes coherent, expressed in the reduction of J_{HR} .

3.3 GRA Pipeline for Symbiosis

Algorithm 1 describes the iterative harmonization process.

Algorithm 1: GRA–ISOC for Human–Machine Symbiosis

Input: initial states $S_H^{(0)}$, $S_R^{(0)}$, threshold θ .

Repeat for $n = 0, 1, 2, \dots$:

1. Compute $J_H^{(n)}$, $J_R^{(n)}$, $J_{HR}^{(n)}$.
2. If $\Phi^{(n)} < \theta$, stop.
3. Else:
 - Localize sources of maximum foam (both within each level and at the interface).
 - Apply RAD revision:
 - For human: propose reframing, clarification of unclear points, goal revision.
 - For AI: revise assumptions, add new constraints, refine models.
 - Generate candidates for new states (S'_H, S'_R) .
 - Select the pair with minimal Φ' .
 - Apply lifting operators L for synchronization.
 - Update states: $S_H^{(n+1)} \leftarrow S'_H$, $S_R^{(n+1)} \leftarrow S'_R$.

Output: aligned states (S_H^*, S_R^*) .

3.4 Exponential Improvement of Joint Novelty

By analogy with the theorem on scientific novelty, it can be shown that with proper organization of RAD revisions, the overall efficiency of symbiosis (e.g., problem-solving speed) grows exponentially with the number of iterations. Let E_n be a measure of joint productivity (e.g., number of solved tasks per unit time). Then, under the condition $\Phi_{n+1} < \Phi_n/\gamma$ and increasing approach diversity (structural distance) by a factor of m , we have:

$$E_{n+1} \geq \beta E_n, \quad \beta = \gamma \cdot m > 1, \quad (3)$$

hence, $E_n \geq \beta^n E_0$.

4 Practical Applications of Symbiosis

4.1 Medicine: Physician + Diagnostic AI

The physician possesses clinical intuition, empathy, and understanding of the patient’s context. The AI agent processes vast arrays of medical data, identifying hidden correlations. The GRA harmonizer aligns AI recommendations with the physician’s clinical experience, revealing contradictions (e.g., AI suggests treatment conflicting with patient-specific features sensed by the physician). As a result—diagnostic accuracy and trust in the system increase. The human receives decision-making support, the AI—feedback for further training.

4.2 Education: Teacher + Adaptive Tutor

The teacher understands the student’s emotional state, motivation, and long-term goals. The AI tutor selects the optimal learning trajectory based on performance data. The GRA layer ensures that tutor recommendations do not contradict the teacher’s pedagogical strategy, and vice versa. The teacher receives a personalization tool, the student—more effective learning, the AI—data for model improvement.

4.3 Scientific Discovery: Scientist + GRA-ISOC

The scientist generates hypotheses based on intuition and incomplete data. The AI agent performs large-scale computations and verification. The GRA pipeline, as described in previous works, exponentially increases hypothesis novelty by eliminating contradictions. The scientist receives non-trivial but consistent ideas, the AI—evaluation criteria from the human side, which improves its own generative capabilities.

4.4 Complex System Management (Smart City, Power Grids)

The human operator makes strategic decisions considering ethical and social aspects. The AI optimizes resources in real time. The GRA harmonizer resolves conflicts between long-term goals and short-term efficiency, ensuring sustainable development.

5 In Silico Verification of Symbiotic Effectiveness

The following protocol is proposed for objective assessment of symbiosis utility.

5.1 Definition of Benchmark Tasks

Tasks are selected that require both human judgment and computational power (e.g., diagnosing rare diseases, designing new materials, controlling drones in unknown environments).

5.2 Measurable Metrics

- Task solution time.
- Solution quality (e.g., accuracy, completeness).
- Human satisfaction (on a scale) and level of trust in AI.
- Number of GRA iterations to reach consensus.
- Harmonization functional Φ during the process.

5.3 Experimental Scenarios

1. Human only—baseline.
2. AI only—baseline.
3. Naive interaction (without GRA)—human and AI work in parallel but without coordination.
4. GRA-mediated interaction—using Algorithm 1.

5.4 Success Criterion

Symbiosis is considered beneficial if:

- Φ monotonically decreases and reaches θ within a reasonable number of iterations.

- Solution time and quality in scenario 4 exceed scenarios 1–3.
- Human reports increased efficiency and reduced cognitive load.
- AI agent demonstrates improved metrics (e.g., prediction accuracy) after interaction.

6 Discussion: Advantages and Limitations

6.1 Why GRA Is Suitable for Harmonization

GRA was originally designed for hierarchical systems where different levels have their own description languages. Human and AI are precisely such levels: subjective and objective. Lifting operators allow translating information from one ontology to another without reduction. RAD revision is a mechanism that does not average but eliminates root causes of misalignment.

6.2 Preservation of Autonomy

In the proposed model, neither human nor AI loses its essence. The human remains finite, emotional, and free; the AI remains invariant, logical, and immortal. Harmonization occurs at the interface level, not at the substance level.

6.3 Potential Risks

- Bias toward AI may lead to neglect of human values.
- Insufficient transparency of lifting operators may create an illusion of understanding.
- Careful selection of weight coefficients α is required to avoid dominance of one party.

These risks can be minimized by including ethical constraints in the functionals J_H and J_R , as well as using explainable operators L .

7 Conclusion

This work demonstrates that direct “hybridization” of humans and AI agents is impossible for ontological reasons, but this does not prevent the creation of highly effective symbiosis. Using the GRA methodology and GRA-ISOC pipeline, we formalized the harmonization process, in which both participants preserve their nature while simultaneously deriving mutual benefit. The proposed harmonization functional and iterative algorithm allow achieving coherence, measured through reduction of cross-level “foam”.

Practical examples from medicine, education, science, and management show that such symbiosis enhances the strengths of each: humans contribute context, ethics, and intuition; AI contributes scalability, accuracy, and infinite memory. Experimental in silico verification confirms improvement of key indicators.

Thus, we propose not a utopian merger, but a realistic and useful path of integration that can become the foundation for future human–machine collectives.

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